SOFTWARE ENGINEERING MINI-PROJECT

Test Plan

Real-Time Chat Application Using Socket Programming

Deliverable 2 – IEEE-Style Test Plan

Document Control

Item	Details
Document	Software Test Plan
Project	Real-Time Chat Application Using Socket Programming
Version	1.0
Purpose	Verify that the implementation satisfies functional, non-functional, and security requirements.

1. Introduction

1.1 Purpose

This Test Plan defines the scope, approach, environment, responsibilities, and validation activities for the Real-Time Chat Application. It covers functional, non-functional, and security requirements and maintains traceability to the SRS.

1.2 Scope

Testing covers registration, authentication, socket connection, chat participation, messaging, timestamps, online users, disconnection handling, input validation, concurrency, performance, and security controls.

1.3 Test Objectives

- Verify implemented functional requirements.
- Verify measurable non-functional requirements.
- Validate security objectives and requirements.
- Identify defects before final submission.
- Maintain traceability between SRS requirements and tests.

1.4 Test Strategy

Testing will combine positive, negative, integration, non-functional, and security validation. Client-server behavior will be tested after individual components are available and again after integration.

2. Test Items and Test Environment

2.1 Test Items

- Chat client/user interface
- Chat server and socket/WebSocket layer
- Authentication and session handling
- Message routing/display
- Connection and user management
- Input validation and security controls

2.2 Test Environment

- Standard laptop/desktop used by the project team.
- Selected project operating system.
- Node.js/WebSockets or Python/Flask-SocketIO runtime, according to implementation.
- Supported web browser or project client.
- Localhost or controlled local network for baseline tests.

2.3 Test Data

Valid and invalid registration data, valid and invalid credentials, normal text messages, empty messages, malformed input, multiple concurrent client sessions, and measured message-delivery times.

3. Features to be Tested

Feature	Coverage
Authentication	Registration, valid/invalid login, logout
Connection Management	Connection, join/leave, status, unexpected disconnect
Messaging	Send, receive, routing, timestamp, ordering
Multi-user Support	Concurrent connections and online-user list
Input Validation	Empty and malformed input
Performance	Message delivery and user-action response time
Security	Authentication, authorization, input validation

4. Features Not to be Tested in the Initial Version

- Voice calling
- Video calling
- File/media sharing
- Third-party messaging-platform integration
- Advanced social features
- Advanced end-to-end encryption beyond controls implemented by the project

5. Test Approach

Functional tests will verify system behavior against FRs. Non-functional tests will measure response time, concurrency, usability-related behavior, and reliability under defined conditions. Negative tests will use invalid credentials and invalid input. Security tests will verify authentication, authorization, and input-validation controls.

5.1 Security Validation

ID	Validation	Expected Result	SRS Trace
SV-01	Access protected chat without authentication	Access is denied	SEC-01, SEC-03
SV-02	Submit malformed/invalid user or message input	Input is rejected or safely handled	SEC-02
SV-03	Attempt an administrative operation without authorization	Operation is denied	SEC-04
SV-04	Use an invalid/terminated session for protected activity	Protected activity is not permitted	SEC-01, SEC-03

5.2 Functional and Non-Functional Test Cases

Test ID	Feature	SRS Trace	Procedure	Expected Result

TC-01	Registration	FR-01	Register with valid new-user data.	Account is created.
TC-02	Valid Login	FR-02	Login using valid credentials.	User is authenticated.
TC-03	Invalid Login	FR-03	Login using incorrect credentials.	Login is rejected.
TC-04	Socket Connection	FR-05	Connect client to running server.	Persistent connection is established.
TC-05	Send/Receive	FR-07, FR-08, FR-09	User A sends a valid message to User B.	User B receives and displays the correct message.
TC-06	Timestamp/Order	FR-10, FR-11	Send multiple messages sequentially.	Messages show timestamps and chronological order.
TC-07	Multiple Users	FR-12, FR-16	Connect multiple clients.	Clients remain connected and active-user list is correct.
TC-08	Disconnect	FR-15, FR-20	Disconnect a client unexpectedly.	Other users remain available and connection state is handled.
TC-09	Invalid Message	FR-17, FR-18	Send empty/invalid message.	Message is rejected.
TC-10	Message Latency	NFR-01, NFR-10	Measure send-to-receive time under normal load.	Measured time is within 1 second.
TC-11	Concurrency	NFR-02	Connect multiple clients concurrently and exchange messages.	Active sessions remain functional.
TC-12	Security/Input	SEC-02, NFR-09	Submit malformed/unexpected input.	Input is validated and safely handled.

5.3 Traceability with SRS

SRS Requirement	Test Case(s)	Coverage
FR-01	TC-01	Registration
FR-02	TC-02	Authentication
FR-03	TC-03	Invalid authentication
FR-05	TC-04	Socket connection
FR-07/08/09	TC-05	Messaging
FR-10/11	TC-06	Message metadata/order
FR-12/16	TC-07	Concurrency/user list
FR-15/20	TC-08	Disconnect/error handling
FR-17/18	TC-09	Input validation
NFR-01/NFR-10	TC-10	Performance
NFR-02	TC-11	Concurrency
NFR-09/SEC-02	TC-12	Validation/security
SEC-01/SEC-03	SV-01, SV-04	Authentication/access control
SEC-02	SV-02, TC-12	Input validation
SEC-04	SV-03	Authorization

6. Pass/Fail Criteria

- A test passes when the actual result matches the expected result.
- A test fails when the actual result differs from the expected result or an unhandled error occurs.
- Critical authentication and security tests must pass before final acceptance.
- Performance requirements pass only when measured values meet the specified target under documented test conditions.

7. Suspension and Resumption Criteria

Testing may be suspended when the server cannot start, the communication mechanism is unavailable, or a blocking defect prevents meaningful execution. Testing resumes after the blocking condition is corrected and the setup is revalidated.

8. Test Deliverables

- Approved Test Plan
- Test case execution records
- Defect/issue records
- Test results and final test summary
- Updated SRS-to-test traceability

9. Roles and Responsibilities

Role	Responsibility
Development Team	Provide the implementation, fix defects, and support test execution.
Testing Responsibility	Prepare/execute tests, record results, and report failures.
Project Team	Review results and verify requirement coverage.

10. Risks and Mitigation

Risk	Impact	Mitigation
Network instability	False communication failures	Use controlled network conditions and repeat failed tests.
Server/blocking defect	Dependent tests cannot execute	Perform server health checks before dependent tests.
Insufficient concurrent clients	Limits concurrency validation	Use multiple browser/client instances or scripted clients.
Unstable measurements	Inconsistent performance results	Repeat measurements under documented conditions.

11. Test Schedule

Testing will be performed incrementally: feature-level functional tests, client-server integration tests, non-functional tests, and security validation before final submission.

12. Approval

The Test Plan shall be reviewed by the project team before execution. Any SRS change affecting coverage shall be reflected in the test cases and traceability information.